

Bhanu Nagpure

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SUMMARY

Android engineer with **2+ years** of hands-on experience building and shipping production apps on Google Play. Deep expertise in **Kotlin, Jetpack Compose, and the Android SDK**, with a proven track record across on-device AI (LiteRT-LM), custom camera pipelines (CameraX), MVVM+Hilt architecture, and Play Store publishing.

TECHNICAL SKILLS

Android: Kotlin, Java, Jetpack Compose (Material 3), Android SDK, MVVM/MVI, Hilt, WorkManager, CameraX
On-Device AI: LiteRT-LM (TFLite runtime), Gemma 2/3, Phi 3.5 Mini, MediaPipe FaceMesh, Google ML Kit
Data & Async: Proto DataStore, Room, Kotlin Coroutines & Flow, Retrofit, REST APIs
Firestore & Monetisation: Firebase (Auth, Crashlytics, Analytics), RevenueCat (IAP), Google AdMob
Tools: Git, GitHub, Android Studio, Gradle (Kotlin DSL), CI/CD, Linux, Google Cloud Platform

EXPERIENCE

Founder & Lead Mobile Engineer

Aug 2025 – Present

Devanshu Studios

Remote

- Shipped **Chakuli** — **Offline AI Chatbot** (Play Store closed beta): privacy-first Android assistant running Gemma 2/3, Phi 3.5 Mini, and SmolLM entirely on-device via **LiteRT-LM** — zero cloud calls for AI inference.
- Built full **Jetpack Compose** (Material 3) UI, **Hilt DI**, **Proto DataStore**, **WorkManager** background model downloads, custom **CameraX** pipeline, Agent Skills system, and encrypted key storage.
- Architected and shipped **PyMaster** — Flutter Python learning platform (50k+ LOC) with Gemini AI Tutor, RevenueCat IAP, and Firebase backend, live on Play Store.

Software Developer Intern

Oct 2022 – May 2024

BETIC-GHRCE

Nagpur, India

- Built an **Android (Kotlin/Java)** app for TMJ disorder screening integrating **MediaPipe FaceMesh** and **Google ML Kit**, improving diagnostic classification accuracy by **60%**.
- Boosted camera performance by **30%** via optimised ML Kit integration enabling real-time facial landmark detection — contributed to a filed patent (Pub. No. 202321022476).

Open Source Contributor

Feb 2025 – Present

Wikimedia Foundation (Scribe-Android)

Remote

- Developed tablet-specific keyboard layouts in **Kotlin** for 7+ languages; resolved UI bugs and refactored the **KeyHandler** module for improved maintainability.

Research Internships

Apr 2024 – Oct 2024

Mitacs (Canada) & Athabasca University

Edmonton / Athabasca, AB, Canada

- Mitacs Globalink: Led Flutter cross-platform (iOS, Android, Web) field-trip app development; presented at the 3rd Annual TRESL Lab Symposium, Canada.
- Athabasca University: Delivered **95% accuracy** text ideology model (Transformers/BERT); built prototype researcher web interface.

PROJECTS

Chakuli — Offline AI Chatbot | Kotlin, Jetpack Compose, LiteRT-LM, Hilt, CameraX

Aug 2025 – Present

- On-device Android AI assistant running **Gemma 2/3, Phi 3.5 Mini, SmolLM** via **LiteRT-LM** — no internet for inference, full data privacy.
- Agent Skills system, encrypted storage (EncryptedSharedPreferences), foreground WorkManager for model downloads, Protobuf DataStore for conversation persistence.

OCANpredict — OSMF Screening App | Kotlin, Java, MediaPipe FaceMesh, Firebase

Dec 2022 – Jul 2023

- Android diagnostic app using **MediaPipe FaceMesh** for real-time OSMF screening, boosting efficiency by **40%**; won multiple national awards and contributed to granted patent (Pub. No. 202321022476).

Autonomous Garbage Collection Robot | Python, YOLOv5, Computer Vision

Jun 2023 – Sep 2023

- Computer vision system (YOLOv5) for autonomous floating waste detection, reducing manual effort by **60%**. Published at 8th ICACS 2024, ACM, Hong Kong.

EDUCATION

G H Raisoni College of Engineering

B.Tech, Computer Science & Engineering | CGPA: 8.7/10.0

Nagpur, India

Dec 2021 – Jun 2025

AWARDS, PUBLICATIONS & PATENTS

Awards: 3rd Place — Avishkar 2024; Runner-up — DIPEX 2023 & SHRISTI 2023; BIREC E-YUVA Grant (Rs. 10 lakh, 2023); Winner — MEDHA & MEDIC-2022.

Publication: “Autonomous Floating Garbage Collection Device using CV and Robotics.” *8th ICACS 2024, ACM*. [\[Link\]](#)

Publication: “V-Safe-Anywhere: Empowering Women’s Safety with Wearable AI and IoT.” *IGI Global*, Mar 2024. [\[Link\]](#)

Patent: Smart Chairside TMJ Examination and Muscle Activity Detector Tool. (Pub. No. 202321022476, Mar 2023 — Granted.)

Medium: Technical blogger on indie Android/Flutter dev — medium.com/@bhanu.r.nagpure.8082